

Azar Louzi

PhD | M2MO | MVA | Centrale Paris

Dual citizenship: France, Morocco. Born on May 8, 1994.

70 rue Nollet, 75017 Paris, France.

+33 7 71 68 81 66 | azar.louzi@lpsm.paris | [azarlouzi.github.io](https://github.com/azarlouzi)

Experience

Vector Institute, McMaster University – Toronto, Ontario, Canada Nov 26-Oct 28
Postdoctoral Fellow. Foundations of interpretable AI reasoning and applications to mathematical finance. *Keywords: AI Theory, Finance*

Sorbonne Université – Paris, France Oct 25-Sep 26
Postdoctoral Researcher. Ergodic convergence rates for a Langevin dynamic with a non-convex potential satisfying the Łojasiewicz property. *Keywords: Langevin, Łojasiewicz*

BNP Paribas CIB – Paris, France Dec 19-Aug 21
Quantitative Analyst. Developed features for FRTB within the pricing library and related systems.

University of Paris Saclay – Paris, France Apr 19-Sep 19
Research Intern. Comparative study in terms of robustness of different calibration methods for the rough Bergomi volatility model.

Octo Technology – Casablanca, Morocco Jul 17-Nov 17
Data Scientist. Consulting and proof of concept services for clients of the company.

Inria CVN – Paris, France Dec 16-Jun 17
Researcher. Analysis of a large scale majorization minimization algorithm.

Inria CVN – Paris, France May 16-Oct 16
Research Intern. Practical and theoretical analysis of a stochastic majorization minimization algorithm.

Education

University of Paris Cité – Paris, France 2021-24
PhD in Applied Mathematics. Multilevel approximation schemes for the value-at-risk and expected shortfall. *Keywords: Risk Measures, Monte Carlo*

University of Paris Cité – Paris, France 2018-19
Masters M2MO (Stochastic Modeling, Finance Track). *With high distinction. Key subjects: Stochastic Calculus, Quantitative Finance*

Ecole Normale Supérieure Paris Saclay – Paris, France 2015-16
Masters MVA (Machine Learning and Computer Vision). *Key subjects: Machine Learning, Deep Learning*

CentraleSupélec (Ecole Centrale Paris) – Paris, France 2013-16
Engineering training with a focus on applied mathematics.

Lycée Louis-le-Grand – Paris, France 2011-13
Preparatory undergraduate education.

Publications

A. Kratsios, [A. Louzi](#), X. Yang. “Dear GD where is the ERM? Estimating the probability that gradient descent fails on a universal attention head”. (Forthcoming.)

[A. Louzi](#). “A weighted Poincaré inequality for semi-log-concave measures”. (Forthcoming.)

[A. Louzi](#), G. Pagès. “Langevin dynamics on Łojasiewicz potentials”. (Forthcoming.)

[A. Louzi](#). “Nested elicibility of the expected shortfall”. (Forthcoming.)

[A. Louzi](#). “Stochastic gradient descent revisited”. 2024. [arXiv:2412.06070](https://arxiv.org/abs/2412.06070).

S. Crépey, N. Frikha, [A. Louzi](#), J. Spence. “Adaptive multilevel stochastic approximation of the value-at-risk”. *Annals of Applied Probability* 36(5), 2026. DOI:10.1214/26-AAP2340.

S. Crépey, N. Frikha, [A. Louzi](#), G. Pagès. “Asymptotic error analysis of multilevel stochastic approximations for the value-at-risk and expected shortfall”. *Electronic Journal of Probability* 29, 2024. DOI:10.1214/24-EJP1246.

S. Crépey, N. Frikha, [A. Louzi](#). “A multilevel stochastic approximation algorithm for value-at-risk and expected shortfall estimation”. *Finance & Stochastics* 29, 2025. DOI:10.1007/s00780-025-00573-5.

Talks & Presentations

Bachelier World Congress 2026. Bologna, Italy. *Talk*.

Séminaire Bachelier Paris 2026. Paris, France. *Talk*.

Séminaire de Probabilités et Mathématiques Financières du LaMME 2026. Paris, France. *Talk*.

Advances in Financial Mathematics 2026. Paris, France. *Talk*.

Groupe de Travail Mathématiques Financières et Actuarielles, Probabilités Numériques du LPSM 2025. Paris, France. *Talk*.

Groupe de Travail Statistiques et Probabilités du LPSM 2025. Paris, France. *Talk*.

Groupe de Travail Méthodes Stochastiques et Finance du CERMICS 2024. Paris, France. *Talk*.

University of Michigan LSA Financial/Actuarial Mathematics Seminars 2024. Detroit, Michigan, USA. *Virtual talk*.

International Conference on Computational Finance 2024. Amsterdam, Netherlands. *Talk*.

Paris Bachelier Doctoral Seminar 2024. Paris, France. *Talk*.

QuantMinds International 2023. London, UK. *Poster*.

International Conference on Stochastic Programming 2023. Davis, California, USA. *Talk*.

Monte Carlo Methods 2023. Paris, France. *Poster*.

MathRisk 2023. Udine, Italy. *Talk*.

Monte Carlo and Quasi-Monte Carlo 2022. Linz, Austria. *Talk*.

Peer Review

Journal of Machine Learning Research.

Teaching

Advanced Mathematics (Tutorial), Bachelor 1, Sorbonne Université, 2026. 48h.

Numerical Analysis (Tutorial), Bachelor 2, Sorbonne Université, 2026. 18h.

Numerical Analysis (Lab), Bachelor 2, Sorbonne Université, 2026. 24h.

Financial Mathematics (Tutorial), Master 1, University of Paris Cité, 2024. 48h.

Office and Internet Tools (Lab), Bachelor 1, University of Paris Cité, 2024. 12h.

Financial Mathematics (Tutorial), Master 1, University of Paris Cité, 2023. 48h.

Probability (Tutorial), Master 1, University of Paris Cité, 2022. 6h.

Prizes & Awards

Natural Sciences and Engineering Research Council of Canada 2026
Canada Impact+ Research Training Award. A one-time initiative designed to recruit international postdoctoral researchers, in priority areas, to Canada.

French Ministry of Higher Education and Research 2025
Qualification for the Section of Applied Mathematics. Qualification allowing to apply for an Assistant Professor position.

Academia Foundation 2013
Excellence Prize. Attributed to Moroccan students who ranked first among their peers at French engineering school entrance exams.

AEFE (Agency for French Education Abroad) 2011-16
Excellence Scholarship. Attributed to best ranking students from overseas French secondary education, to allow them to pursue their higher education in France.

French Secondary Education – Morocco 2010
First Prize. Mathematical olympiads.

Skills

Languages *English* fluent; *French*, *Arabic* bilingual; *Spanish* advanced (DELE B2)

Computer Science C++, Ada, Go, Python, R, Matlab, LaTeX, Git, Linux, Vim

Hobbies Chess, Photography, Illustration